

GRAE-3DMOT: Geometry Relation-Aware Encoder for Online 3D Multi-Object Tracking

Hyunseop Kim¹ Hyo-Jun Lee² Yonguk Lee³
 Jinu Lee³ Hanul Kim⁴ Yeong Jun Koh^{1*}

¹Chungnam National University ²Kangwon National University

³42dot Inc. ⁴Seoul National University of Science and Technology

hyunseop95@gmail.com, gywns6298@kangwon.ac.kr, yonguk.lee@42dot.ai

jinu.lee@42dot.ai, hukim@seoultech.ac.kr, yjkoh@cnu.ac.kr

Abstract

Recently, 3D multi-object tracking (MOT) has widely adopted the standard tracking-by-detection paradigm, which solves the association problem between detections and tracks. Many tracking-by-detection approaches establish constrained relationships between detections and tracks using a distance threshold to reduce confusion during association. However, this approach does not effectively and comprehensively utilize the information regarding objects due to the constraints of the distance threshold. In this paper, we propose GRAE-3DMOT, Geometry Relation-Aware Encoder 3D Multi-Object Tracking, which contains a geometric relation-aware encoder to produce informative features for association. The geometric relation-aware encoder consists of three components: a spatial relation-aware encoder, a spatiotemporal relation-aware encoder, and a distance-aware feature fusion layer. The spatial relation-aware encoder effectively aggregates detection features by comprehensively exploiting as many detections as possible. The spatiotemporal relation-aware encoder provides spatiotemporal relation-aware features by combining spatial and temporal relation features, where the spatiotemporal relation-aware features are transformed into association scores for MOT. The distance-aware feature fusion layer is integrated into both encoders to enhance the relation features of physically proximate objects. Experimental results demonstrate that the proposed GRAE-3DMOT outperforms the state-of-the-art on the nuScenes. Our approach achieves 73.7% and 70.2% AMOTA on the nuScenes validation and test sets using CenterPoint detections. Code is available at <https://github.com/alitkddhfcjs/GRAE-3DMOT>.

*Corresponding author

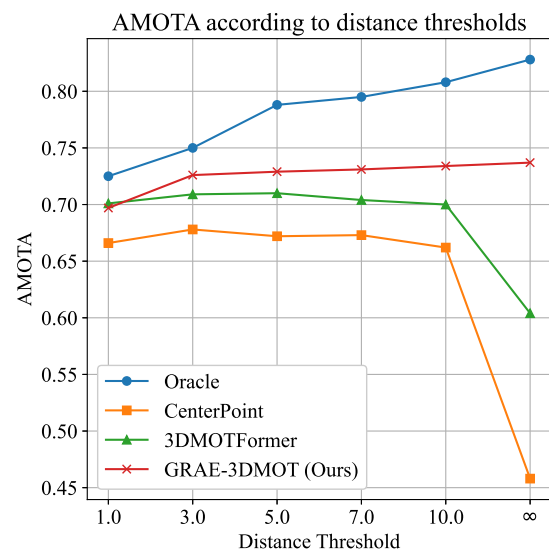


Figure 1. The 3D MOT performance (AMOTA) of oracle, 3DMOTFormer, CenterPoint, and the proposed GRAE-3DMOT according to distance thresholds for connecting edges in graphs with detections of CenterPoint. We design oracle to match tracks and detections using ground-truth identity with distance threshold constraints.

1. Introduction

3D multiple object tracking (MOT) is a crucial task in many applications, such as autonomous driving and mobile robots. It involves localization and identification of surrounding objects over time, making accurate and consistent online tracking essential for fundamental tasks in these applications. Recent advancements in 3D MOT are due to the strong performance of high-performance object detection within a tracking-by-detection framework. State-of-the-art 3D MOT algorithms [9, 14, 16, 27, 28, 31] utilize detection results from off-the-shelf 3D object detectors to associate

instances across frames. Many tracking-by-detection mannered algorithms [1, 7, 27] use non-learned algorithms such as the Kalman filter [10] with geometric association metrics based on center distance or 3D IoU. However, these methods require heuristic designs and handcrafting.

Recently, learning-based methods [15, 20, 35] have emerged for end-to-end tracking, also known as tracking-with-query. Inspired by DETR [6], these methods use transformer queries to predict object sequences and extend detected queries to track objects in subsequent frames. However, balancing detection and tracking performance remains challenging, often leading to low tracking performance due to suboptimal detection. As a result, the learning-based tracking-by-detection approach is still popular for 3D MOT. Some methods use temporal object detection models with accurate velocity regression [19, 25, 31] or neural networks to learn appearance models [11, 23, 28, 34]. While powerful, these methods require training for specific environments and still involve manual parameter tuning and heuristic design for association. Other approaches [9, 12, 28] introduce Graph Neural Networks (GNNs) to learn object tracking using only geometric cues. These networks construct a graph with each node representing an instance feature, and at each frame, associate existing tracked objects with new detections using a bipartite graph. However, some methods [12, 28] require a fixed time window to evolve the graph, while others [9] use dynamic bipartite graphs. Despite this, they remain graph-dependent, limiting their ability to associate unconnected nodes.

Figure 1 shows the 3D MOT performance (AMOTA) according to distance thresholds for connecting edges with detections of CenterPoint [31]. In the oracle, which matches tracks and detections using ground truth identity, AMOTA improves as the distance threshold increases. This indicates that the same object may be positioned at significantly different locations across adjacent frames and it is beneficial to connect nodes as many as possible. However, 3DMOTFormer [9] and CenterPoint [31] experience the performance degradation as the threshold increases, due to their inability to effectively establish associations among the increasingly extensive set of nodes.

In this work, we propose a geometry-based 3D MOT method, called GRAE-3DMOT. The proposed method generates features by exploiting the relationships between objects, differing from existing methods [9, 12, 28] that only utilize features between connected nodes with a fixed class-agnostic distance threshold. Instead, our method builds a graph with an infinite distance threshold to comprehensively analyze as many relationships between objects as possible. It allows for a sufficient number of association candidates even when detections are physically distant from tracks, which yields the state-of-the-art as in Figure 1.

We introduce the geometry relation-aware encoder,

which consists of three components. First, the spatial relation-aware encoder generates spatial relation-aware features based on a spatial relation graph, capturing relative information between all detections in the current frame to enable effective feature aggregation. Second, the spatiotemporal relation-aware encoder produces temporal relation features from a temporal relation graph, capturing relative information between all detections and tracks. It then combines spatial and temporal relation features to obtain spatiotemporal relation-aware features, which are used for associating detections with tracks. Third, the distance-aware feature fusion layer implicitly updates object features based on their distance, emphasizing the relation features of physically close objects. Finally, we estimate association scores between tracks and detections using features processed by the geometry relation-aware encoder and propagate these track features frame-by-frame for long-term interaction.

Experimental results demonstrate that the proposed GRAE-3DMOT outperforms the state-of-the-arts on the nuScenes [5] tracking benchmark. With CenterPoint detections, the proposed method achieves 73.7% and 70.2% AMOTA on the validation and test sets, respectively, yielding the state-of-the-art performance among all geometry-based approaches.

In summary, the contributions are as follows:

- The proposed GRAE-3DMOT establishes relationships between objects without distance constraints.
- We introduce the spatial relation-aware encoder, utilizing the spatial relation graph to exploit relative information between objects, and the spatiotemporal relation-aware encoder, which leverages the temporal relation graph exploiting relative information between detections and tracks.
- GRAE-3DMOT achieves outstanding performance on the nuScenes benchmark with various detectors.

2. Related Work

3D multiple object tracking (MOT) has garnered significant attention due to its crucial role in applications such as autonomous driving and mobile robotics. Numerous approaches have been proposed to enhance tracking accuracy and robustness, addressing challenges such as occlusions, data association, and real-time processing. This section reviews some of the notable methods and advancements in the field. **Tracking-by-Detection** frameworks utilize detection results from off-the-shelf 3D object detectors to associate instances across frames. This method leverages the strong performance of modern object detection algorithms and relies heavily on geometric association metrics. Some approaches focus on identifying objects across a sequence of frames by continuously tracking them, primarily using the predicted track motion based on a Kalman

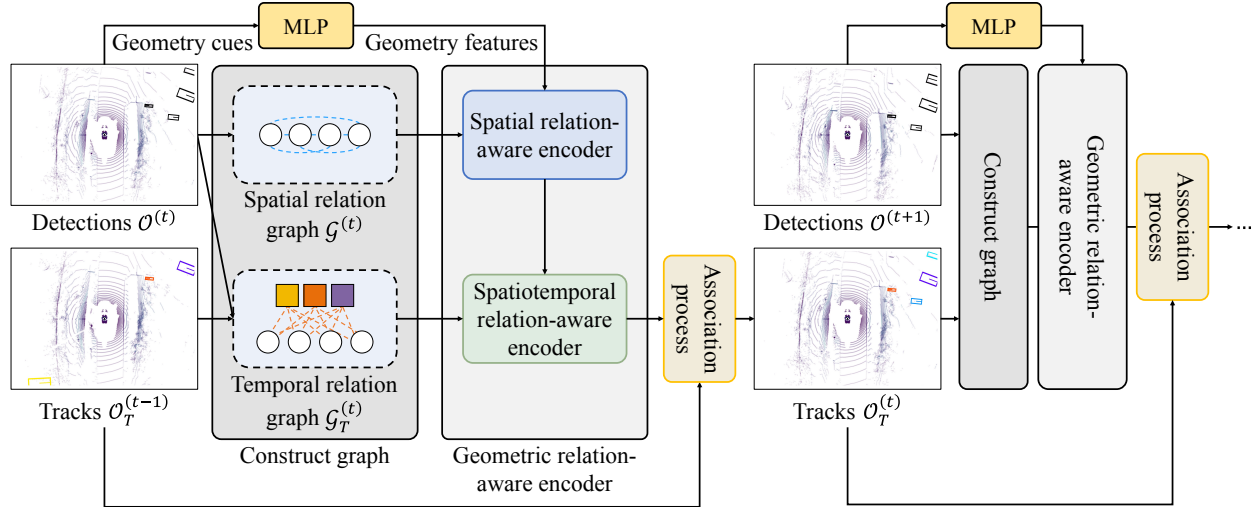


Figure 2. An overview of the proposed GRAE-3DMOT framework. Our model consists of two graphs: a spatial relation graph constructed from the detections in the current frame, and a temporal relation graph constructed from the detections in the current frame and the tracks in the previous frame. The GRAE-3DMOT consists of the geometry relation-aware encoder, which has three components: the spatial relation-aware encoder generates spatial relation-aware features based on a spatial relation graph, the spatiotemporal relation-aware encoder produces temporal relation features from a temporal relation graph, and distance-aware feature fusion layer implicitly updates object features based on their distance.

filter [4, 19, 27]. Other studies improve track management by leveraging detection scores [1, 37] or using appearance matching [29]. These approaches integrate a re-identification module that extracts distinctive appearance features of objects [2, 36]. Various approaches are also being developed to enhance motion through learning, such as utilizing multi-category trajectories [14] and matching learned motion and appearance features [7]. Recent approaches have introduced learning-based methods to address the limitations of non-learned algorithms. For example, models such as SimpleTrack [19] and CenterPoint [31] incorporate accurate velocity regression, while others like GNN3DMOT [28] and CamoMOT [23] utilize neural networks to learn appearance models. Additionally, some approaches integrate Graph Neural Networks (GNNs) to learn object tracking given geometric cues [12, 16]. However, these methods can be graph-dependent, limiting their ability to effectively associate unconnected nodes. **Tracking with Query** leverages specific queries to focus on particular objects of interest, enabling end-to-end detection and tracking. Inspired by DETR [6], these methods utilize track queries to predict object sequences. It primarily employs detected queries and tracked queries [8, 18, 33], storing these queries in a queue and refining them [20]. Another approach projects and tracks objects across different camera views using 3D-to-2D queries [35]. Various methods have been developed for using queries in tracking systems. Some studies differentiate between object queries and track queries [15], while others use perspective detection queries [30] to enhance tracking performance. Additionally,

research is being conducted on methods that use only the 3D motion of tracks [9]. However, balancing detection and tracking performance remains challenging in these frameworks. The reliance on high detection performance can lead to suboptimal tracking results, prompting continued popularity for learning-based tracking-by-detection methods.

3. Proposed methods

Figure 2 shows an overview of the proposed GRAE-3DMOT framework. In this work, we introduce the geometric relation-aware encoder, which has three components. First, the spatial relation-aware encoder produces spatial relation-aware features by utilizing the relative information between detections within the current frame. The second component is the spatiotemporal relation-aware encoder, which explores the relative information between detections and tracks to determine the associations between them. Third, the distance-aware feature fusion layer (DFFL) enhances the relation features of objects that are physically close to each other, thereby emphasizing their proximity-based interactions.

3.1. 3D Object Detection

The proposed network can be easily incorporated with the existing 3D object detectors [25, 30, 31]. In this paper, we perform 3D MOT experiments with three different detection models, BEVFormerv2 [30], StreamPETR [25] and CenterPoint [31]. For each frame $I^{(t)}$, the 3D object detector provides the set of object detections $\mathcal{O}^{(t)} = \{o_1^{(t)}, \dots, o_{N_t}^{(t)}\}$, where each detection $o_i^{(t)}$ contains geometry attributes $\mathbf{g}_i^{(t)}$.

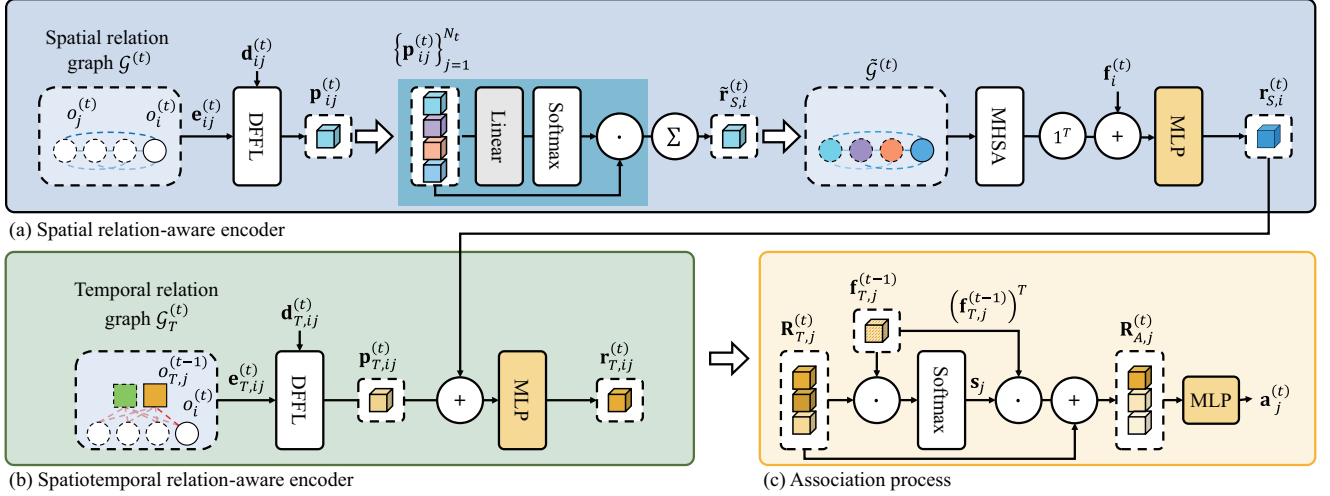


Figure 3. A diagram of the geometric relation-aware encoder. (a) is the spatial relation-aware encoder. (b) is the spatiotemporal relation-aware encoder. (c) is the association process.

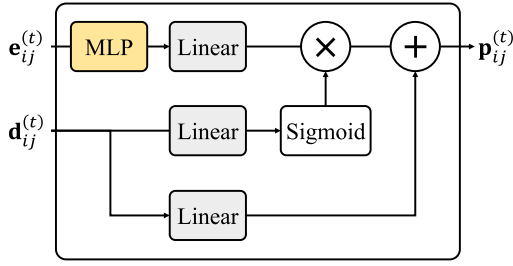


Figure 4. A diagram of the distance-aware feature fusion layer. Here, $\mathbf{g}_i^{(t)}$ includes box center positions $x_i^{(t)}, y_i^{(t)}, z_i^{(t)}$, sizes $w_i^{(t)}, l_i^{(t)}, h_i^{(t)}$, rotations, one-hot encoded class, and a detection score. The 3D object detector provides a velocity $(\Delta x_i^{(t)}, \Delta y_i^{(t)})$ to the next frame for each detection, but we do not use the velocity as the geometry attributes. We project each geometry cue $\mathbf{g}_i^{(t)}$ into a geometry feature $\mathbf{f}_i^{(t)} \in \mathbb{R}^C$ through multi-layer perceptron (MLP), where C is a feature dimension.

3.2. Geometric Relation-Aware Encoder

The geometric relation-aware encoder is composed of the spatial relation-aware encoder and the spatiotemporal relation-aware encoder. The output features of the geometric relation-aware encoder are used to yield the association between detections and tracks. Additionally, these output features are used to update the features of the tracks.

Spatial relation-aware encoder. Figure 3(a) shows a diagram of the spatial relation-aware encoder. The spatial relation-aware encoder generates enriched detection features at the current frame t by aggregating features from all detections, based on relative information between detections. Given the detection set $\mathcal{O}^{(t)}$, we form a spatial relation

graph $\mathcal{G}^{(t)} = (\mathcal{V}^{(t)}, \mathcal{E}^{(t)})$, where the node set $\mathcal{V}^{(t)}$ consists of the object set $\mathcal{O}^{(t)}$ and each node $o_i^{(t)}$ is connected to the other ones with edges in the edge set $\mathcal{E}^{(t)}$. An edge $\mathbf{e}_{ij}^{(t)}$ from $o_i^{(t)}$ to $o_j^{(t)}$ is defined as the difference between geometry attributes, which is given by

$$\mathbf{e}_{ij}^{(t)} = \begin{cases} \mathbf{g}_j^{(t)} - \mathbf{g}_i^{(t)} & \text{if } \|\mathbf{d}_{ij}^{(t)}\|_2 < d_S, \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

where $\mathbf{d}_{ij}^{(t)} = [x_j^{(t)} - x_i^{(t)}, y_j^{(t)} - y_i^{(t)}]$ denotes the center distance vector from $o_i^{(t)}$ to $o_j^{(t)}$ and d_S is a threshold of center distance between detections. In this work, we set $d_S = \infty$ to conduct a comprehensive analysis of as many relationships between detections as possible.

For each edge $\mathbf{e}_{ij}^{(t)}$, we perform the distance-aware feature fusion layer (DFFL) in Figure 4 to obtain a distance-aware spatial relation prior

$$\mathbf{p}_{ij}^{(t)} = \psi(\mathbf{e}_{ij}^{(t)}, \mathbf{d}_{ij}^{(t)}) = \text{Sigmoid}(\text{Linear}(\mathbf{d}_{ij}^{(t)})) \cdot \text{Linear}(\text{MLP}(\mathbf{e}_{ij}^{(t)})) + \text{Linear}(\mathbf{d}_{ij}^{(t)}), \quad (2)$$

where $\psi(\cdot, \cdot)$ denotes the DFFL function. Thus, based on the center distance $\mathbf{d}_{ij}^{(t)}$, DFFL can emphasize the relation prior that are physically close to the detection $o_i^{(t)}$. After computing $\{\mathbf{p}_{ij}^{(t)}\}_{j=1}^{N_t}$ for all edges of $o_i^{(t)}$, we combine the spatial relation priors to compute a spatial relation-aware feature $\tilde{\mathbf{r}}_{S,i}^{(t)}$ for $o_i^{(t)}$ respect to the other detections in $\mathcal{G}^{(t)}$. Then, the spatial relation-aware feature $\tilde{\mathbf{r}}_{S,i}^{(t)}$ is defined as

$$\tilde{\mathbf{r}}_{S,i}^{(t)} = \sum_{j=1}^{N_t} \text{Softmax}(\text{Linear}(\mathbf{p}_{ij}^{(t)})) \cdot \mathbf{p}_{ij}^{(t)}. \quad (3)$$

We repeat the process to compute the spatial relation-aware features for all detections $\{\tilde{\mathbf{r}}_{S,i}^{(t)}\}_{i=1}^{N_t}$. We then further enhance them through the relation-aware attention, which is designed to achieve more comprehensive interactions between detections. We build a new graph $\tilde{\mathcal{G}}^{(t)} = (\mathcal{V}^{(t)}, \tilde{\mathcal{E}}^{(t)})$ with the same node to $\mathcal{G}^{(t)}$, but the different edge set $\tilde{\mathcal{E}}^{(t)}$, which is computed using the spatial relation-aware features $\{\tilde{\mathbf{r}}_{S,i}^{(t)}\}_{i=1}^{N_t}$ instead of $\{\mathbf{g}_i^{(t)}\}_{i=1}^{N_t}$. Thus, an edge from $o_i^{(t)}$ to $o_j^{(t)}$ is defined as the differences between $\tilde{\mathbf{r}}_{S,i}^{(t)}$ and $\tilde{\mathbf{r}}_{S,j}^{(t)}$ if $\|\mathbf{d}_{ij}^{(t)}\|_2 < d_S$. Then, enhanced features are derived by stacking edges in $\tilde{\mathcal{E}}$ for each detection $o_i^{(t)}$, and the stacked ones are processed by multi-head self-attention (MHSA) to produce the enhanced spatial relation features $\mathbf{R}_{S,i}^{(t)} \in \mathbb{R}^{N_t \times C}$. Finally, we aggregate the enhanced features in $\mathbf{R}_{S,i}^{(t)}$ to obtain a spatial relation-aware feature $\mathbf{r}_{S,i}^{(t)}$, which is given by

$$\mathbf{r}_{S,i}^{(t)} = \text{MLP}((\mathbf{1}^T \mathbf{R}_{S,i}^{(t)})^T + \mathbf{f}_i^{(t)}) \quad (4)$$

where $\mathbf{1}$ denotes a vector of ones. Here, the geometry feature $\mathbf{f}_i^{(t)}$ is also combined for reliable aggregation.

Spatiotemporal relation-aware encoder. We design the spatiotemporal relation-aware encoder to leverage the relative information between detections at the current frame and tracks at the previous frame. Figure 3(b) shows a diagram of the spatiotemporal relation-aware encoder. We build a temporal relation graph $\mathcal{G}_T^{(t)} = (\mathcal{V}^{(t)}, \mathcal{V}_T^{(t-1)}, \mathcal{E}_T^{(t)})$, which is a bipartite graph connecting nodes in $\mathcal{V}^{(t)}$ and nodes in $\mathcal{V}_T^{(t-1)}$ with edges in $\mathcal{E}_T^{(t)}$. $\mathcal{V}^{(t)}$ is correspond to $\mathcal{O}^{(t)}$, while $\mathcal{V}_T^{(t-1)}$ includes the set of tracks at the previous frame $t-1$, $\mathcal{O}_T^{(t-1)} = \{o_{T,1}^{(t-1)}, o_{T,2}^{(t-1)}, \dots, o_{T,M_{t-1}}^{(t-1)}\}$, where M_{t-1} denotes the number of tracks at $t-1$ th frame. Here, each track $o_{T,j}^{(t-1)}$ contains a track feature $\mathbf{f}_{T,j}^{(t-1)} \in \mathbb{R}^C$, which is sequentially updated as the MOT process progresses.

Similar to (1), an edge $\mathbf{e}_{T,ij}^{(t)}$ from $o_i^{(t)}$ to $o_{T,j}^{(t-1)}$ is computed by difference of geometry attributes, which given by

$$\mathbf{e}_{T,ij}^{(t)} = \begin{cases} \mathbf{g}_j^{(t-1)} - \mathbf{g}_i^{(t)} & \text{if } \|\mathbf{d}_{T,ij}^{(t)}\|_2 < d_T, \\ 0 & \text{otherwise} \end{cases} \quad (5)$$

where $\mathbf{d}_{T,ij}^{(t)} = [x_j^{(t-1)} + \Delta x_j^{(t-1)} - x_i^{(t)}, y_j^{(t-1)} + \Delta y_j^{(t-1)} - y_i]$ denotes the center distance vector from $o_i^{(t)}$ to $o_{T,j}^{(t-1)}$. Here, the center positions for track is compensated with the estimated velocity, which the object detector provides, to obtain the reliable edge. d_T is a threshold of center distance between detections and track, which is set to $d_T = \infty$ in this work. This approach not only conducts the comprehensive analysis of as many relationships as possible between detections and tracks, but also allows for a larger number of

matching between detections with tracks without distance constraints.

Given edges in $\mathcal{E}_T^{(t)}$, we obtain the distance-aware temporal relation prior $\mathbf{p}_{T,ij}^{(t)} = \psi(\mathbf{e}_{T,ij}^{(t)}, \mathbf{d}_{T,ij}^{(t)})$ using the DFFL function. Then, we compute a spatiotemporal relation-aware feature $\mathbf{r}_{T,ij}^{(t)} \in \mathbb{R}^C$ by combining the temporal relation prior and the spatial relation-aware feature, which is given by

$$\mathbf{r}_{T,ij}^{(t)} = \text{MLP}(\mathbf{p}_{T,ij}^{(t)} + \mathbf{r}_{S,i}^{(t)}). \quad (6)$$

For each track $o_{T,j}^{(t-1)}$, after computing $\{\mathbf{r}_{T,ij}^{(t)}\}_{i=1}^{N_t}$ for all detections, we stack them to compose spatiotemporal relation-aware features $\mathbf{R}_{T,j}^{(t)} \in \mathbb{R}^{N_t \times C}$.

3.3. Association Process

As shown in the Fig 3(c), we compute association features $\mathbf{R}_{A,j}^{(t)} \in \mathbb{R}^{N_t \times C}$, whose i th row vector denotes an association feature between a track $o_{T,j}^{(t-1)}$ and a detection $o_i^{(t)}$, by aggregating the spatiotemporal relation-aware features $\mathbf{R}_{T,j}^{(t)}$ and the track feature $\mathbf{f}_{T,j}^{(t-1)}$. Specifically, we distribute the track feature $\mathbf{f}_{T,j}^{(t-1)}$ into the spatiotemporal relation-aware features based on affinity scores $\mathbf{s}_j \in \mathbb{R}^{N_t}$ between the track and detections. The affinity scores are computed by

$$\mathbf{s}_j = \text{Softmax}(\mathbf{R}_{T,j}^{(t)} \cdot \mathbf{f}_{T,j}^{(t-1)}) \quad (7)$$

whose i th element denotes the affinity score between $o_{T,j}^{(t-1)}$ and $o_i^{(t)}$. Then, the association features $\mathbf{R}_{A,j}^{(t)} \in \mathbb{R}^{N_t \times C}$ are defined as

$$\mathbf{R}_{A,j}^{(t)} = \text{MLP}(\mathbf{R}_{T,j}^{(t)} + \mathbf{s}_j \cdot (\mathbf{f}_{T,j}^{(t-1)})^T). \quad (8)$$

Finally, the association features $\mathbf{R}_{A,j}^{(t)}$ are transformed, through MLP, into association scores $\mathbf{a}_j^{(t)} = \text{MLP}(\mathbf{R}_{A,j}^{(t)}) \in \mathbb{R}^{N_t}$ between the track $o_{T,j}^{(t-1)}$ and all detections $\{o_i^{(t)}\}_{i=1}^{N_t}$. After computing association scores $\{\mathbf{a}_j^{(t)}\}_{j=1}^{M_{t-1}}$ for all tracks, we compose an association score matrix $\mathbf{A}^{(t)} = [\mathbf{a}_1^{(t)}, \mathbf{a}_2^{(t)}, \dots, \mathbf{a}_{M_{t-1}}^{(t)}]$.

We perform Hungarian matching [13] on the association score matrix $\mathbf{A}^{(t)}$ to achieve overall association between detections in $\mathcal{O}^{(t)}$ and tracks in $\mathcal{O}_T^{(t-1)}$. Here, we discard the matching pair with affinity scores lower than β . Then, based on the matching pairs, we update the tracks $\mathcal{O}_T^{(t-1)}$ to $\mathcal{O}_T^{(t)}$ by adding the matched objects in $\mathcal{O}^{(t)}$. Specifically, when a detection $o_i^{(t)}$ is matched to a track $o_{T,j}^{(t-1)}$, $o_i^{(t)}$ becomes $o_{T,j}^{(t)}$ with its geometry attributes. Also, the track feature is updated to $\mathbf{f}_{T,j}^{(t)}$ by sampling i th association feature from $\mathbf{R}_{A,j}^{(t)}$ followed by MLP. After the association, there may be unmatched tracked objects as well as unmatched detected

	AMOTA↑	AMOTP↓	MOTA↑	MT↑	ML↓	TP↑	FP↓	FN↓	IDS↓	FRAG↓
OGR3MOT [32]	0.656	0.620	0.554	5278	2094	95264	17877	24013	288	371
PolarMOT † [12]	0.664	0.566	0.561	5701	1686	97909	17856	21414	242	332
CenterPoint [31]	0.638	0.555	0.537	5584	1681	95877	18612	22928	760	529
CBMOT [1]	0.649	0.592	0.545	5319	1966	94916	16469	24092	557	450
SimpleTrack ‡ [19]	0.668	0.550	0.566	5476	1780	95539	17514	23451	575	591
ImmortalTracker [24]	0.677	0.599	0.572	5565	1669	97584	18012	21661	320	477
GNN-PMB [16]	0.678	0.560	0.563	5698	1622	97274	17071	21521	770	431
3DMOTFormer [9]	0.682	0.496	0.556	5466	1894	95790	18322	23337	438	529
GRAE-3DMOT (Ours)	0.702	0.475	0.574	5785	1781	99023	19343	20323	219	366

Table 1. Results on the test set in nuScenes using CenterPoint [31] detections. AMOTA and AMOTP are the primary metrics. The best results are boldfaced. †denotes offline methods and ‡denotes methods using 10Hz data.

	AMOTA↑	AMOTP↓	MOTA↑	IDS↓	FRAG↓
OGR3MOT [32]	0.693	0.627	0.602	262	332
PolarMOT † [12]	0.711	-	-	213	332
CenterPoint [31]	0.665	0.567	0.562	562	424
CBMOT [1]	0.675	-	-	494	-
SimpleTrack ‡ [19]	0.696	0.547	0.602	-	403
ImmortalTracker [24]	0.702	-	0.601	-	385
GNN-PMB [16]	0.707	0.560	-	650	345
3DMOTFormer [9]	0.712	0.515	0.607	341	436
GRAE-3DMOT (Ours)	0.737	0.488	0.636	150	315

Table 2. Results on the validation set in nuScenes using CenterPoint [31] detections. The best results are boldfaced. † denotes offline methods and ‡ denotes methods using 10Hz data.

objects. Remained detected objects are regarded as newly appearing objects, which are used to initialize new trajectories. Also, unmatched tracked objects record the number of successive unmatched frames τ , which is a count-based track management. Unmatched tracks are permanently removed when τ is higher than λ frames.

4. Experiments

4.1. Datasets and metrics

We evaluate the performance of GRAE-3DMOT on a large-scale 3D MOT datasets: nuScenes [5]. NuScenes contains 700, 150, and 150 scenes for training, validation, and test sets, respectively. NuScenes is collected using multiple sensors, including multiview cameras, a 32-beam LiDAR, and RADARs. The dataset provides keyframes annotated at a frequency of 2 Hz in each sequence. NuScenes provides a subset of 7 mobile categories, such as cars, pedestrians, bicycles, buses, motorcycles, trailers, and trucks, for tracking.

We assess the performance following the official evaluation metrics for multi-object tracking tasks from nuScenes. The main metrics [26] are average multi-object tracking accuracy (AMOTA) and average multi-object tracking precision (AMOTP). In addition, nuScenes employs a variety of secondary metrics, such as MOTA, MOTP, IDS, and FRAG from CLEAR MOT [3].

4.2. Implementation details

3D object detector. For 3D object detectors as the geometry attributes, we first use the lidar-based detector, CenterPoint [31], which is mainly used in the geometry-based state-of-the-art (SOTA) [9]. Furthermore, we use two recent multiview-based detectors BEVFormerV2 [30] and StreamPETR [25] to validate the generalization of GRAE-3DMOT. We apply non-maximum suppression (NMS) with a 3D IoU threshold of 0.1 to filter out duplicate detections.

Training. We use the Focal loss [21] to compute an association loss $\mathcal{L}_a$ between $\mathbf{A}^{(t)}$ and the ground-truth association for each frame. We set mini-sequences to 8 adjacent frames from training scenes, corresponding to 2.5s at a frequency of 2Hz. We use AdamW optimizer [17] for training 21 epochs with a batch size of 8 on Nvidia GeForce 4090Ti GPU. We use a learning rate of 2e-04 and a weight decay of 0.01. Hyperparameters C , β , and λ are set to 128, 0.1 and 12, respectively.

4.3. Main Results

Comparative assessment. Table 1 shows the comparison of the proposed GRAE-3DMOT with the existing geometry-based 3D MOT methods on the test set in nuScenes dataset. For the fair comparisons, all methods in Table 1 uses CenterPoint [31] to obtain geometry attributes. We observe that the proposed method significantly outperforms other methods for AMOTA, AMOTP, MT, TP, and FN metrics. Notice that the proposed method surpasses the state-of-the-art 3DMOTFormer [9] by margins of 2.0 and 2.1 in terms of AMOTA and AMOTP, respectively. Also, the best MT, ML, TP, and FN scores indicate that the proposed method is effective and robust to maintain accurate tracks over a long time. Also, GRAE-3DMOT achieves the best score on MOTA. Table 2 compares GRAE-3DMOT with 3D MOT methods on the validation set. The proposed method achieves the best performance in all evaluation metrics. Especially, compared to 3DMOTFormer [9], GRAE-3DMOT improves AMOTA and AMOTP scores by 2.4 and 2.1, respectively.

Detector (Backbone)	mAP↑	NDS↑	Tracker	AMOTA↑	AMOTP↓	MOTA↑	IDS↓	FRAG↓
BEVFormerv2 [30] (ResNet-50)	46.0	44.7	CenterPoint [31]	0.413	1.028	0.388	789	983
			3DMOTFormer [9]	0.469	1.044	0.409	568	909
			GRAE-3DMOT (Ours)	0.486	1.068	0.428	387	883
StreamPETR [25] (VoV-99)	48.2	57.1	CenterPoint [31]	0.464	1.025	0.421	778	959
			3DMOTFormer [9]	0.506	1.039	0.443	425	859
			GRAE-3DMOT (Ours)	0.521	1.029	0.452	299	794

Table 3. Comparison of GRAE-3DMOT with other methods using various detectors on the validation set. The best results are boldfaced.

Training	Inference	AMOTA↑	AMOTP↓	MOTA↑	IDS↓
CenterPoint [31]	StreamPETR [25]	0.518	1.030	0.455	381
	BEVFormerv2 [30]	0.481	1.064	0.422	362
StreamPETR [25]	CenterPoint [31]	0.732	0.496	0.630	227
	BEVFormerv2 [30]	0.472	1.066	0.420	451

Table 4. Generalization across detectors. The proposed method was training on source detector. We compared this trained the proposed method on detections from another target detector.

	FPS↑	Parameter (MB)↓	FLOPs (MB)↓
3DMOTFormer [9]	47.24	0.74	1.46
GRAE-3DMOT (Ours)	56.19	0.57	0.97

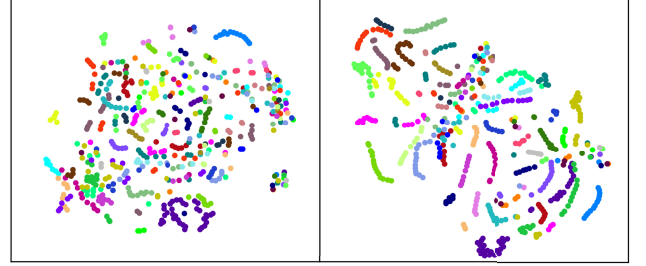
Table 5. Efficiency comparison of GRAE-3DMOT with 3DMOTFormer.

d_S	d_T	AMOTA↑	AMOTP↓	MOTA↑	IDS↓	FRAG↓
3.0	3.0	0.727	0.496	0.628	345	289
5.0	5.0	0.729	0.496	0.630	272	284
5.0	∞	0.732	0.495	0.632	200	309
∞	5.0	0.730	0.495	0.631	270	284
∞	∞	0.737	0.488	0.636	150	315

Table 6. Ablation study for the graph representation according to the thresholds d_S and d_T .

Various detectors. In this work, to validate the generalization capability of GRAE-3DMOT in the presence of the inaccurate geometry information, we additionally employ vision-centric detectors, BEVFormerv2 [30] and StreamPETR [25], resulting in the lower detection performance than CenterPoint [31], which achieves a performance of mAP 68.5 and NDS 61.5. In Table 3, we observe that the proposed GRAE-3DMOT outperforms 3DMOTFormer [9] for all detectors with significant AMOTA improvements, which demonstrates the generalization capability of GRAE-3DMOT. Also, we conduct domain generalization experiments across different 3D detectors in training and inference. By comparing scores in Tables 3 and 4, we observe that there is only slight performance degradation compared to the standard setting (same detector in training and inference). This indicates that the superior domain generalization ability of GRAE-3DMOT.

Efficiency evaluation. Table 5 compares GRAE-3DMOT with 3DMOTFormer for FPS, parameters, and FLOPs. FPS



(a) $d_S = 5.0$, $d_T = 5.0$

(b) $d_S = \infty$, $d_T = \infty$

Figure 5. Comparison of t-SNE visualization for track features over 20 frames according to the edge thresholds. (a) Visualization of $d_S = 5.0$ and $d_T = 5.0$. (b) Visualization of $d_S = \infty$ and $d_T = \infty$. Features belonging to the same track are represented with the same color.

$\mathbf{e}_{ij}^{(t)}$	AMOTA↑	AMOTP↓	MOTA↑	IDS↓	FRAG↓
$\mathbf{g}_j^{(t)}$	0.725	0.494	0.630	277	359
$\mathbf{f}_j^{(t)} - \mathbf{f}_i^{(t)}$	0.729	0.493	0.631	208	357
$\mathbf{g}_j^{(t)} - \mathbf{g}_i^{(t)}$	0.737	0.488	0.636	150	315

Table 7. Ablation study for the edge representation.

for GRAE-3DMOT and 3DMOTFormer are measured on Nvidia GeForce 4090Ti GPU and the other metrics are from the official source code. The proposed model surpasses 3DMOTFormer for all metrics, which validates the efficiency of the proposed GRAE-3DMOT.

4.4. Ablation Study

We conduct various ablation studies using the CenterPoint detector [31] on the validation set in nuScenes [5] to demonstrate the effectiveness of the proposed components.

Graph representation. The proposed method uses two types of graphs: spatial and temporal relation graphs. Unlike the existing graph-based methods [9, 12, 32] use distance thresholds to truncate nodes in the graph, all nodes are connected in the proposed relation graph without any truncation. Table 6 shows the performance of GRAE-3DMOT according to the center distance thresholds d_S for the spatial relation graph and d_T for the temporal relation graph. We observe that the performance improves as the both thresholds increase. Especially, GRAE-3DMOT with thresholds $d_S = \infty$ and $d_T = \infty$ provides the best performance, which

	Spatial relation-aware encoder	Spatiotemporal relation-aware encoder	AMOTA↑	AMOTP↓	MOTA↑	IDS↓	FRAG↓	TP↑	FP↓	FN↓
M1	✓		0.720	0.502	0.620	609	214	82335	12870	18953
M2		✓	0.729	0.494	0.629	245	307	84410	14195	17242
M3	✓	✓	0.737	0.488	0.636	150	315	84104	13152	17643

Table 8. Ablation study for the geometric relation-aware encoder.

	AMOTA↑	AMOTP↓	MOTA↑	IDS↓	FRAG↓
3DMOTFormer	0.729	0.494	0.633	213	289
w/o MHSA	0.732	0.495	0.631	294	291
Proposed	0.737	0.488	0.636	150	315

Table 9. Ablation study for detection feature aggregation.

indicates that aggregating features from as many detections as possible can provide informative features for 3D MOT. Also, $d_T = \infty$ allows for a sufficient number of association candidates, even when detections are physically distant from tracks, resulting in the high performance.

To analysis the qualitative results of features according to thresholds d_S and d_T , we employ t-SNE [22] as illustrated in Figure 5, which compares t-SNE visualization of track features over 20 frames when the thresholds are set to 5 and ∞ . Here, the features belonging to the same track are represented with the same color. With the threshold 5, even the features of the same track are scattered. On the other hand, with the infinity threshold, consistent and enhanced features are obtained by exploiting relationships as many detections and tracks as possible.

Edge representation. We evaluate four different variants of edge representation for spatial relation graph construction: (1) $\mathbf{e}_{ij}^{(t)} = \mathbf{g}_j^{(t)}$, (2) $\mathbf{e}_{ij}^{(t)} = \mathbf{f}_j^{(t)} - \mathbf{f}_i^{(t)}$, and (3) $\mathbf{e}_{ij}^{(t)} = \mathbf{g}_j^{(t)} - \mathbf{g}_i^{(t)}$ (proposed setting). In Table 7, we observe that the edge representation with simple geometry attribute $\mathbf{e}_{ij}^{(t)} = \mathbf{g}_j^{(t)}$ degrades AMOTA score by 1.2 compared to the proposed approach, which uses the geometry difference. This indicates that initializing edges based on geometry differences is effective for feature aggregation. Also, we see that differences of geometry attributes are more effective than those of embedded features for edge representation.

Geometric Relation-Aware Encoder. In Table 8, we analyze the effectiveness of the spatial and spatiotemporal relation-aware encoders. M1 uses only the spatial relation-aware encoder. In M1, the spatial relation-aware features are used instead of spatiotemporal relation-aware features. M2 uses only the temporal relation-aware encoder, where the geometry features $\{\mathbf{f}_i^{(t)}\}_{i=1}^{N_t}$ are used instead of the spatial relation-aware feature $\{\mathbf{r}_{S,i}^{(t)}\}_{i=1}^{N_t}$. We see that M3 (proposed method) yields the best performance with the significant improvements on all metrics compared to M1 and M2.

Detection feature aggregation. In many graph-based 3D MOT methods [9, 12, 28], detection feature aggregation is

	AMOTA↑	AMOTP↓	MOTA↑	IDS↓	FRAG↓
w/o DFFL	0.729	0.495	0.631	208	357
with DFFL	0.737	0.488	0.636	150	315

Table 10. Ablation study for distance-aware feature fusion layer (DFFL).

generally performed to obtain enhanced detection features. For this purpose, we also propose the spatial relation-aware encoder as the detection feature aggregation. Table 9 shows the ablation study for detection feature aggregation. First, we replace the proposed spatial relation-aware encoder with the graph transformer encoder in 3DMOTFormer [9], which is designed to aggregate detection features. In Table 9, we observe that the proposed feature aggregation method outperforms 3DMOTFormer, demonstrating the effectiveness of the spatial relation-aware encoder. Second, we perform the spatial relation-aware encoder without multi-head self-attention (MHSA), resulting in the degraded performance. It indicates that the repetitive graph construction in the spatial relation-aware encoder is essential for accurate results.

Distance-aware feature fusion layer. We conduct an ablation study on the distance-aware feature fusion layer (DFFL) as shown in Table 10. We observe that AMOTA and AMOTP decreases by 0.8 and 0.7 without DFFL. This validates the effectiveness of DFFL to interfuse distance information into spatial relation features.

5. Conclusions

In this paper, we proposed the novel 3D online multi-object tracking called GRAE-3DMOT that only relies on geometry cues. GRAE-3DMOT aggregates features by comprehensively exploiting the relationships between objects with the infinite distance threshold, differing from existing methods that only utilize features between connected nodes with fixed distance thresholds. GRAE-3DMOT effectively combines spatial relation and temporal relation features to yield spatiotemporal relation-aware features. Under the infinite distance threshold for the temporal graph, GRAE-3DMOT effectively associates detections and tracks using the informative spatiotemporal relation-aware features. The proposed GRAE-3DMOT achieved the state-of-the-art performance on the nuScenes dataset, outperforming all other geometric-based data association approaches with various detectors.

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